

Lecture 6: Firms and Production (Chapters 3–4)

Applied Microeconomics — Master Notes · by Jakob V. Stangel

Required reading: Friberg, Ch. 3 — From technology to costs (production functions, costs, isoquants & cost minimization) · Ch. 4 — Supply by price-taking firms (profit, $MR = MC$, shutdown, long-run supply). Lecturer: L. Butera — deck "**Lecture 6: Production.**" (The lecture is titled "Chapter 4 / Firms & production"; the production-and-cost machinery is actually Friberg Ch. 3, and the profit/supply half is Ch. 4 — this note covers both.)

Résumé. Lectures 4–5 were the **consumer**; now it's the **firm**. The firm is the mirror image of the consumer: where the consumer maximised **utility** subject to a **budget**, the firm **minimises cost** subject to a **production function**, and then **maximises profit**. The lecture runs the whole chain **technology** → **cost** → **supply**: build the **production function** $Q = f(K, L)$; derive **short-run product** (marginal product, diminishing returns) and **cost curves** (MC, ATC, AVC, AFC); add **revenue** to get the **profit-maximising rule** $MR = MC = p$ and the firm's **supply curve**; add the **shutdown rule**; let **free entry** drive the **long run** to **zero profit**; and finally solve the **optimal input mix** (L^*, K^*) with **isoquants** and **isocosts** (the exact analogue of the consumer's indifference-curve optimum).

★ The heart of the lecture — the firm as a two-stage optimiser.

- **Stage 1 — cost minimisation (how to produce):** for any target output, pick the input mix where the **isoquant** is **tangent to the lowest isocost** — $MRTS = w/r$, i.e. the **last-dollar rule** $MP_L/w = MP_K/r$. This gives the **cost function** $C(q)$.
- **Stage 2 — profit maximisation (how much to produce):** produce where **marginal revenue = marginal cost**. A **price-taking firm** has $MR = p$, so the rule is $p = MC$ — and the firm's **supply curve** is its **MC curve** (above min AVC).
- **The time dimension:** in the **short run** capital is **fixed** (so there are fixed costs, and a shutdown decision); in the **long run** everything is **variable** and **free entry drives economic profit to zero** ($p = MC = \min ATC$).

Key concepts / "modes" to use: perfect competition & the price taker · production function $Q = f(K, L)$; Cobb-Douglas $Q = \sqrt{KL}$ · short run vs long run (K fixed vs all variable) · marginal product $MP_L = \partial Q / \partial L$; law of diminishing marginal returns · returns to scale (CRS · IRS · DRS); the exponent-sum shortcut · FC · VC · TC; ATC · AVC · AFC; $MC = dTC/dq$; MC cuts ATC & AVC at their minima · $TR = pq$; $\pi = TR - TC$; $MR = dTR/dq$ · the output rule $MR = MC = p$; the firm's supply curve · the shutdown rule ($p < \min AVC$) · free entry → long-run zero economic profit; opportunity cost & economic rent · isoquants · $MRTS = MP_L/MP_K$ · isocost $wL + rK = C$ · the tangency / last-dollar rule.

The firm, and the perfectly-competitive benchmark

Core: the whole lecture lives in **perfect competition**, whose assumptions make each firm a **price taker** — it can sell as much as it likes at the market price **p**, but nothing above it.

- **The five assumptions:** (1) a very large number of firms (no one firm moves the price); (2) **identical (homogeneous)** products; (3) **complete, symmetric information**; (4) **negligible transaction costs**; (5) **free entry and exit** (in the long run).
- **The consequence** — the firm's demand curve is **horizontal** at **p**. Raise your price a cent and you sell **zero** (buyers switch to an identical rival); so the firm **takes p as given**. (*Why nearly flat and not perfectly flat? The firm faces the market's residual demand; with many rivals that residual is hugely elastic — soybean example: market demand is only moderately elastic, yet one farm faces a "super-elastic," almost flat demand.*)
- **The firm** = the decision-maker that chooses **what** and **how** to produce; modelled as a single actor with one objective — **maximise profit** (and, as a sub-step, **minimise cost**).

Inputs and the production function

Core: production turns **inputs** into **output**. We simplify the firm's many inputs to two: **capital K** and **labour L**.

- **Factors of production:** **labour** (human time/effort), **capital** (things *used but not used up* — buildings, machines, servers; includes **intangible** capital like patents), plus **land** and **raw/intermediate materials**.
- **The production function** gives the maximum output from given inputs:

$$Q = f(K, L).$$

The lecture's running form is **Cobb-Douglas**, e.g. $Q = \sqrt{KL} = K^{1/2} L^{1/2}$. It can also be written as a **table** (output for each K-row $\times$ L-column).

Short run vs long run (timing matters)

Core: the distinction is about **what can be varied**, not the calendar.

	Short run (SR)	Long run (LR)
Inputs	at least one is fixed — we assume capital K is fixed	all inputs variable
Costs	fixed costs exist (rent, machines)	all costs variable
Example	a sandwich shop can hire an extra pair of hands fast, but can't add an oven overnight	it can build a whole new sandwich station

In plain terms: the SR is "stuck with your factory"; the LR is "you can rebuild the factory (or leave the industry)."

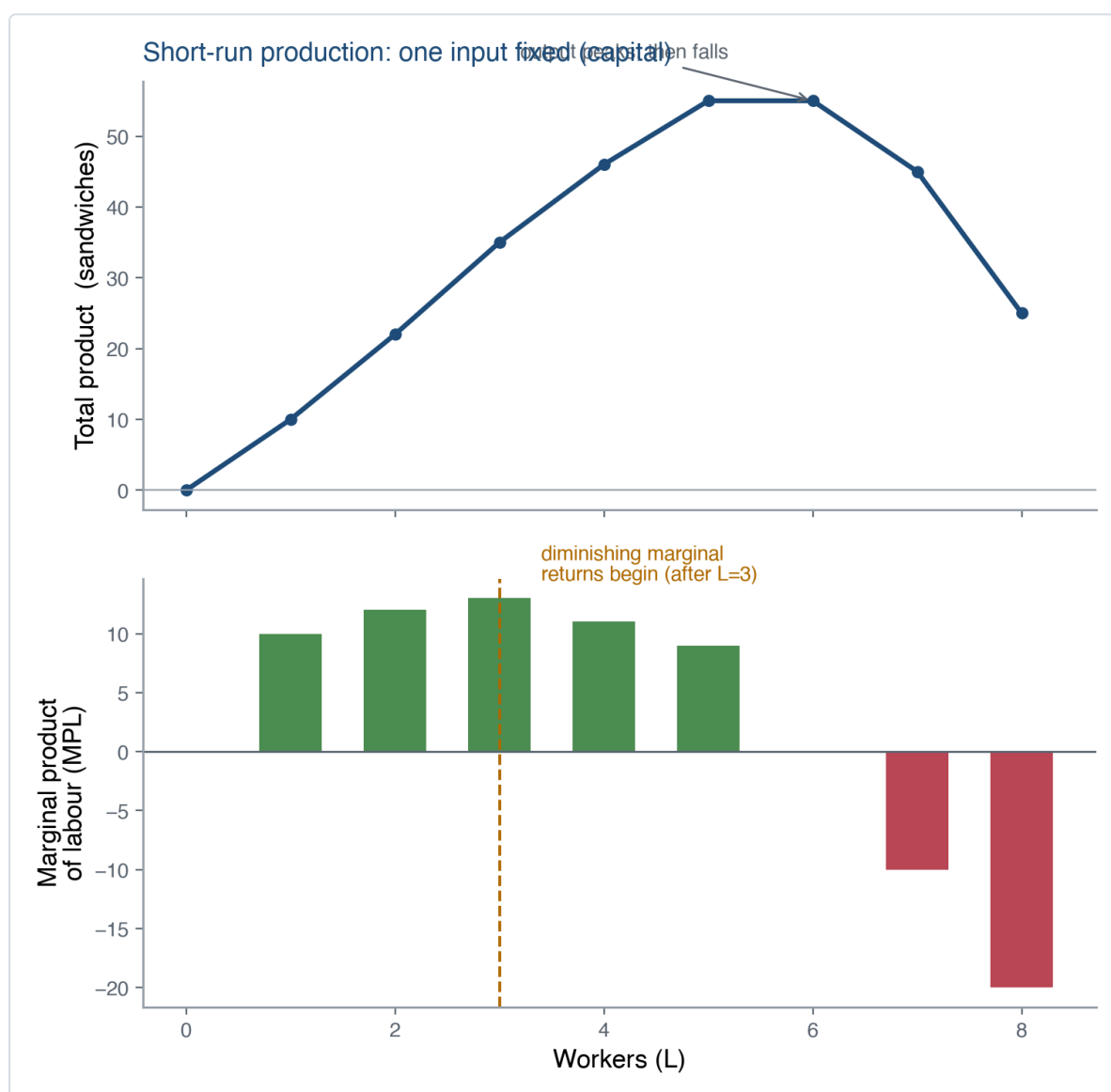
★ Short-run production: marginal product & diminishing returns

Core: with **K fixed**, keep adding **labour** and output rises — but each extra worker eventually adds **less** than the last.

- **Marginal product of labour (MP_L)** = the extra output from one more unit of labour, holding K fixed = the **partial derivative**:

$$MP_L = \partial Q / \partial L. \text{ For } Q = \sqrt{KL}: MP_L = \frac{1}{2}\sqrt{K/L}.$$

- **The law of diminishing marginal returns:** beyond some point, adding more of a **variable** input to **fixed** inputs makes **MP_L fall**. (Too many cooks crowd the one kitchen.)
- **Crucial distinction** — **diminishing marginal product** $\neq$ **falling output**. Total output is still **rising**; it is the **increments** that shrink (and only once MP_L goes **negative** does total output actually fall).



How to read it. **Top:** total product climbs steeply at first, then flattens and finally **turns down** once extra workers get in each other's way. **Bottom:** the **marginal** product (one worker's *addition*) **rises** through the first three workers, then **diminishes** (the amber line), and goes **negative** past the point where total

product peaks. *The onset of diminishing marginal returns is where the marginal-product bars start shrinking — here, after the 3rd worker.*

Returns to scale (a long-run, all-inputs question)

Core: scale all inputs up by the same factor α and ask what happens to **output**.

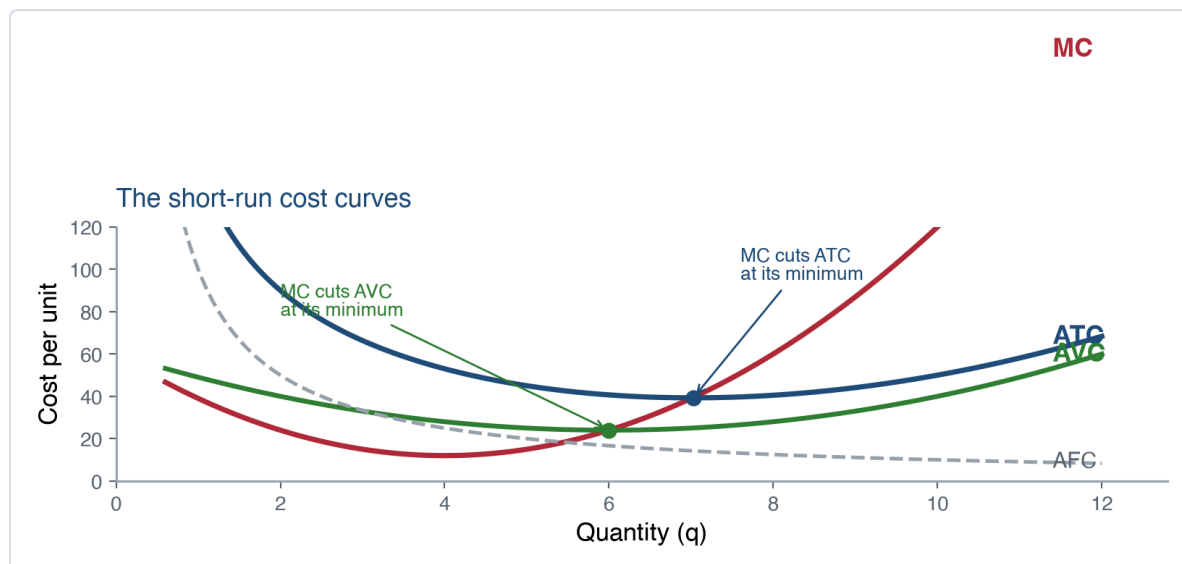
	Definition (double all inputs $\Rightarrow$...)	Formula	Why
Constant returns (CRS)	output doubles	$f(\alpha K, \alpha L) = \alpha \cdot f(K, L)$	replication
Increasing returns (IRS)	output more than doubles	$f(\alpha K, \alpha L) > \alpha \cdot f(K, L)$	specialisation; one big plant beats two small
Decreasing returns (DRS)	output less than doubles	$f(\alpha K, \alpha L) < \alpha \cdot f(K, L)$	coordination gets harder as the firm grows

- **Worked checks:** $Q = \sqrt{KL} \rightarrow f(\alpha K, \alpha L) = (\alpha K)^{1/2} (\alpha L)^{1/2} = \alpha \cdot \sqrt{KL} \rightarrow \text{CRS}$. $Q = K^2 L^2 \rightarrow (\alpha K)^2 (\alpha L)^2 = \alpha^4 \cdot K^2 L^2 \rightarrow \text{IRS}$.
- **Cobb-Douglas shortcut:** sum the exponents. Sum = 1 $\rightarrow$ CRS; > 1 $\rightarrow$ IRS; < 1 $\rightarrow$ DRS.
- (Don't confuse with *economies of scale*, a cost idea: falling **long-run ATC** as output grows — linked to IRS but not identical.)

★ The short-run cost curves

Core: total cost splits into a part that **varies** with output and a part that **doesn't**, and dividing by Q gives the per-unit curves.

- **Total cost** $TC = VC + FC$ — **variable cost** (labour, materials, electricity: rises with Q) + **fixed cost** (rent, equipment: paid regardless in the SR).
- **Per-unit costs** (divide by Q): $ATC = AVC + AFC$, where $ATC = TC/Q$, $AVC = VC/Q$, $AFC = FC/Q$.
- **Marginal cost** — the cost of **one more unit**:
 $MC = \Delta TC / \Delta Q = dTC/dq$ (the derivative of total cost).
- The shapes (and the one fact that always shows up on exams):
 - **AFC** falls continuously (a fixed cost spread over more and more units).
 - **ATC and AVC are U-shaped:** spreading fixed cost pulls ATC **down**, rising MC pulls it **up** $\rightarrow$ a U.
 - ★ **MC cuts ATC and AVC at their minimum points.** *Intuition (the height analogy):* a new unit **cheaper** than the current average pulls the average **down**; **dearer** than the average pulls it **up** — so marginal crosses average exactly at the average's lowest point.



How to read it (illustrative diagram). **MC** (red) slopes up. **AVC** (green) and **ATC** (blue) are U-shaped, with **ATC** sitting above **AVC** by the falling **AFC** (grey dashed) gap. The two marked dots are the key fact: **MC** passes exactly through the bottom of **AVC** and the bottom of **ATC**. Left of each minimum, **MC** is below the average and drags it down; right of it, **MC** is above and pushes it up.

The rewards of doing business: revenue & profit

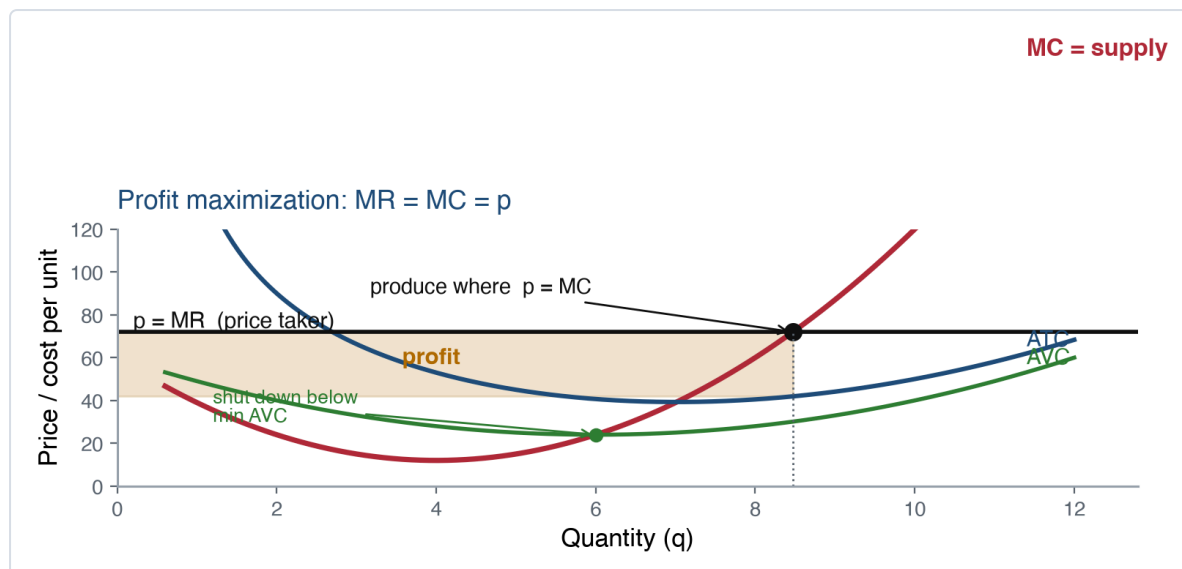
Core: the other half of the firm's problem — what it earns.

- Total revenue $TR = p \times Q$.
- Profit $\pi = TR - TC$.
- Marginal revenue — the revenue from one more unit sold: $MR = \Delta TR / \Delta Q = dTR/dq$. For a price taker, every extra unit sells at the same market price, so $MR = p$.

★ Profit maximisation: $MR = MC = p$

Core: keep producing an extra unit as long as it brings in more than it costs — stop when the last unit's revenue equals its cost.

- The output rule (three equivalent statements): (1) profit is maximised; (2) marginal profit is zero; (3) $MR = MC$. Formally, maximise $\pi(q) = TR(q) - TC(q)$: set $d\pi/dq = MR - MC = 0 \rightarrow MR = MC$.
- For a price-taking firm $MR = p$, so the rule collapses to $p = MC \rightarrow$ and therefore the firm's supply curve is its marginal-cost curve (the upward-sloping part, above min AVC).
- Worked example (Friberg): $C(q) = 10 + q^2$, price $p = 12$. Maximise $12q - (10 + q^2)$: FOC $12 - 2q = 0 \rightarrow q^* = 6$. Profit $\pi = (p - ATC) \cdot q = (12 - 46/6) \cdot 6 = 26 > 0$. (At $p = 10 \rightarrow q^* = 5$; at $p = 14 \rightarrow q^* = 7$ — tracing out the MC/supply curve.)



How to read it (illustrative diagram). The flat black line is $p = MR$ (the price taker's "demand"). The firm slides out along its MC curve and stops where $p = MC$ (the dot). **Profit** is the shaded box: height $(p - ATC)$ times width q^* . If the price line sits **below min AVC** (the green dot), the firm can't even cover its variable costs and **shuts down**.

The shutdown rule (short run)

Core: a firm losing money should keep going **only if producing loses less than closing**.

- In the SR, **fixed costs are unavoidable (sunk)** — you pay the rent whether you open or not. So the firm compares **revenue to variable cost**, not total cost.
- **Shutdown rule:** shut down if $price < \text{minimum AVC}$ (i.e. if $pq < VC$ — revenue can't even cover variable costs). If $p \geq \text{min AVC}$, keep producing: the excess $pq - VC$ ("variable profit") at least **reduces** the loss on fixed costs.
- So the firm's SR **supply curve** = MC above **min AVC**, and **zero** below it. (Shutdown is a **SR** decision; in the LR all costs are avoidable, so a loss-maker simply **exits**.)

Long-run competition: free entry drives profit to zero

Core: the same rule $MR = MC$ still picks output, but now **free entry and exit** pin down the **price**.

- In the LR **all costs are variable** (no fixed cost; $ATC = AVC$) and **firms can enter or exit**. Positive economic profit **attracts entry**; entry **shifts market supply out** and **pushes the price down**. Entry stops only when **profit = 0**.
- **Long-run equilibrium condition:** price is driven to the **minimum of ATC**, where $p = MR = MC = \text{min ATC}$, and **economic profit = 0**.

The LR market **supply curve** is therefore **flat at min ATC** (with identical firms, free entry and constant input prices).

- **"Zero profit"** means zero *economic* profit. **Economic cost = accounting cost + opportunity cost**; at the zero-profit point every input (including the owner's time and capital) earns exactly its **next-**

best return — the firm is doing fine, just not *better* than its alternatives. A return to owning a **scarce** input (prime Champagne land) is an **economic rent**, and can make the LR supply curve slope **up**.
(*Learning-by-doing* — *chips, aircraft* — can even make it slope **down**.)

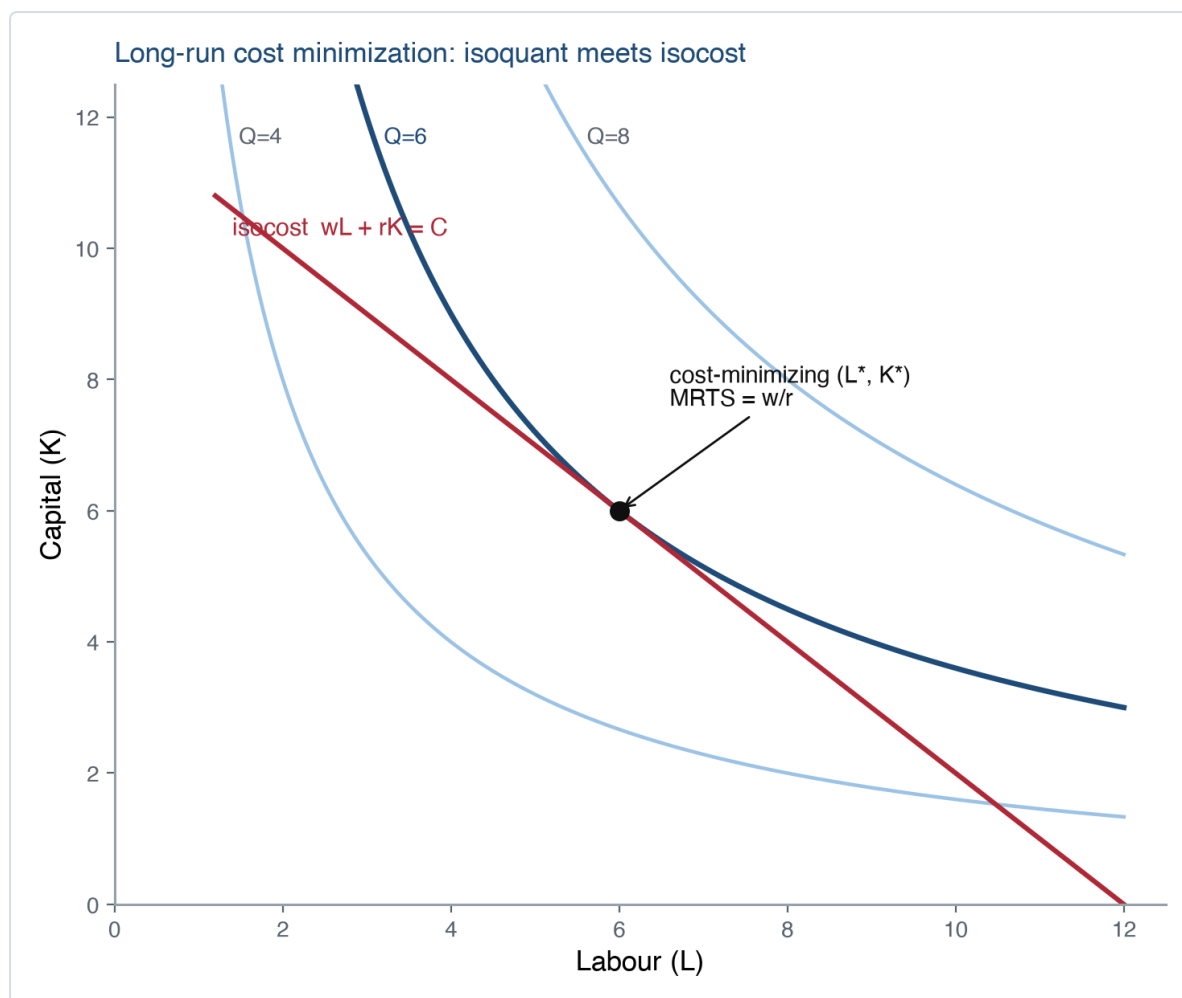
★ Optimal input choice: isoquants, isocosts & cost minimisation

Core: how to produce a given output most cheaply — the exact analogue of the consumer's optimum, with **isoquants** playing the role of indifference curves and **isocosts** the role of the budget line.

- **Isoquant** — all **(L, K)** combinations that make the same output **Q** ("iso" = equal). Shapes mirror the consumer's: **perfect substitutes** → straight line; **perfect complements** → L-shaped (rickshaws need exactly one driver each); the usual **imperfect substitutes** → **convex**, bowed toward the origin.
- **The slope of an isoquant = the marginal rate of technical substitution (MRTS)** — how much **K** you can drop when you add one **L**, holding output constant:

$MRTS = MP_L / MP_K$ (derived by totally differentiating $Q = f(K, L)$ and setting $dQ = 0$).

- **Isocost line** — all input combinations costing the **same total**, $C = wL + rK$ (w = wage, r = rental rate of capital). Rearranged: $K = C/r - (w/r) \cdot L$ → **intercept C/r , slope $-w/r$** .
- **Cost minimisation (three equivalent rules):**
 1. **Lowest-isocost rule:** pick the input bundle where the **lowest isocost** still touches the target isoquant.
 2. **Tangency rule:** where the **isoquant is tangent to the isocost** → slopes equal → **$MRTS = w/r$** .
 3. **Last-dollar rule:** $MP_L/w = MP_K/r$ — the last dollar spent on each input buys the **same extra output**. (*The same algebra, via the Lagrangian: $MP_L/w = MP_K/r = \lambda$.*)
- **Mapping each target Q to its minimised cost gives the cost function $C(q)$** — closing the loop back to the cost curves above.



How to read it (illustrative diagram). Each blue curve is an **isoquant** — a fixed output level, more toward the top-right = more output. The red line is an **isocost** (all input mixes costing the same C). To make $Q = 6$ as cheaply as possible, slide to the **lowest** isocost that still touches that isoquant — the **tangency** point (L^*, K^*) , where the isoquant's slope ($MRTS = MP_L/MP_K$) equals the isocost's slope (w/r).

- **Comparative statics — why factories differ by country.** Same technology, same r , but **Japan** has a **high wage** → a **steeper** isocost → the tangency uses **more capital, less labour** (automation). **Bangladesh** has a **low wage** → **flatter** isocost → **more labour, less capital** (labour-intensive garments). Relative **input prices** pick the technique.

Cases & examples

- **The CPH sandwich shop** — a new office opens next door: in the **SR** you hire hands (variable), but can't add ovens (fixed) → diminishing marginal product; in the **LR** you build a new station.
- **The Wisconsin Cheeseman** — "why not 500 boxes? why not 1500?" → the firm settles at **MR = MC**.
- **Japan vs Bangladesh** — same technology, different wages → capital-intensive vs labour-intensive input mix (comparative statics of cost minimisation).
- **The electricity supply curve** — the market supply stacks sources by MC ("merit order": wind/hydro cheapest, then nuclear, then fossil), so it is flat at low output and steep at the top.
- **Champagne land** — the forgone grape revenue is the **opportunity cost** of any other use; the premium for scarce land is **economic rent**.

- **The Black Death** — plague cut the workforce → each survivor had more land/capital → **MP_L and real wages rose** (marginal product drives wages).

Formula sheet

Concept	Formula
Production function	$Q = f(K, L)$; Cobb-Douglas $Q = K^a L^b$
Marginal product of labour	$MP_L = \partial Q / \partial L$ (for $Q = \sqrt{KL}$: $\frac{1}{2}\sqrt{K/L}$)
Returns to scale	$f(aK, aL) = a \cdot f \Rightarrow$ CRS; $> a \cdot f \Rightarrow$ IRS; $< a \cdot f \Rightarrow$ DRS (Cobb-Douglas: $a + b$ vs 1)
Total / average costs	$TC = VC + FC$; $ATC = TC/Q$; $AVC = VC/Q$; $AFC = FC/Q$
Marginal cost	$MC = dTC/dq$
MC–AC relation	MC cuts ATC and AVC at their minima
Revenue, profit	$TR = pQ$; $\pi = TR - TC$; $MR = dTR/dq$
Output rule	$MR = MC$; price taker $\Rightarrow$ $p = MC$ (supply = MC above min AVC)
Shutdown (SR)	produce iff $p \geq \min AVC$
Long-run equilibrium	$p = MC = \min ATC$, economic profit = 0
Isocost	$C = wL + rK \Rightarrow K = C/r - (w/r)L$
MRTS (isoquant slope)	$MRTS = MP_L / MP_K$
Cost-min (tangency / last dollar)	$MRTS = w/r \Leftrightarrow$ $MP_L / w = MP_K / r$

★ Practice problems (with solutions)

Problem 1 — short-run firm (the lecture's example). A firm has $TC = 2q^2 + 50q + 50$ and faces a market price $p = 100$. (a) Find FC, VC, AFC, AVC, ATC and MC. (b) How much does the firm produce, and what profit does it make?

Solution. (a) $FC = 50$; $VC = 2q^2 + 50q$; $AFC = 50/q$; $AVC = 2q + 50$; $ATC = 2q + 50 + 50/q$; $MC = dTC/dq = 4q + 50$. (b) Price taker $\Rightarrow p = MC$: $100 = 4q + 50 \rightarrow q^* = 12.5$. Then $TR = 100(12.5) = 1250$; $TC = 2(12.5^2) + 50(12.5) + 50 = 312.5 + 625 + 50 = 987.5 \rightarrow \pi = 1250 - 987.5 = 262.5 > 0$. (Since $p = 100 > \min AVC$, the firm produces.)

Problem 2 — long run with free entry. Same cost function $TC = 2q^2 + 50q + 50$; market demand $p = 210 - Q$. Find the **long-run** price, each firm's output, the number of firms, and each firm's profit.

Solution. Free entry drives price to **min ATC**. Minimise $ATC = 2q + 50 + 50/q$: $dATC/dq = 2 - 50/q^2 = 0 \rightarrow q^2 = 25 \rightarrow q^* = 5$, giving **min ATC** = $2(5) + 50 + 50/5 = 70$. So the **LR price** = **70**. Market quantity from demand: $Q = 210 - 70 = 140$. Number of firms $n = Q / q^* = 140 / 5 = 28$. Each firm's profit $\pi = (p - ATC) \cdot q = (70 - 70) \cdot 5 = 0$ — zero economic profit, as required. (*Entry has pushed the price down from 100 to 70 and wiped out the short-run profit.*)

Problem 3 — returns to scale & cost minimisation. (a) Does $Q = K^{1/3} L^{1/2}$ show increasing, constant or decreasing returns to scale? (b) A firm produces with $Q = \sqrt{KL}$, wage $w = 4$, rental rate $r = 1$. In what proportion should it use L and K ?

Solution. (a) Sum of exponents = $1/3 + 1/2 = 5/6 < 1 \rightarrow$ **decreasing returns to scale**. (b) $MP_L = \frac{1}{2}\sqrt{K/L}$, $MP_K = \frac{1}{2}\sqrt{L/K} \rightarrow MRTS = MP_L/MP_K = K/L$. Tangency: $MRTS = w/r \rightarrow K/L = 4/1 \rightarrow K = 4L$. The firm uses **four units of capital per unit of labour** (capital-intensive, because labour is relatively dear).

★ Fun facts & memorable details (Lecture 6: Firms & Production)

Sticky bits from the production lecture.

- **The firm is a consumer in a mirror.** Indifference curve $\leftrightarrow$ **isoquant**; budget line $\leftrightarrow$ **isocost**; $MRS =$ price ratio $\leftrightarrow$ **$MRTS = w/r$** . Learn one optimum, you've learned both.
- **Marginal cuts average at the bottom — always.** Whether it's cost curves or your grade-point average, the **marginal** value crosses the **average** exactly at the average's minimum. Add something below your average and it drops; above, it rises.
- **You can rationally run at a loss.** In the short run, as long as price covers **average variable cost**, staying open loses **less** than closing — because the rent is sunk either way. The shutdown line is **min AVC**, not **min ATC**.
- **"Zero profit" is a success, not a failure.** Long-run zero *economic* profit means every input — including the owner — is earning exactly its best alternative return. It only looks like zero because the **opportunity cost** is already counted.
- **The Black Death raised wages.** Kill a third of the workforce and each survivor works more land $\rightarrow$ higher **marginal product** $\rightarrow$ higher real wages. A grim demonstration that **wages track marginal product**.
- **Why Japanese factories are robotic and Bangladeshi ones aren't.** Same technology — different **wages**. High wages tilt the isocost toward **capital**; low wages toward **labour**. The technique follows the price of the input.
- **One big plant can beat two small ones.** That's **increasing returns to scale** (specialisation) — and, on the cost side, the **economies of scale** that let a few firms dominate aircraft, pharma and chips.